Background

Why monitor moths?

The EU has decided that member states must begin monitoring pollinator diversity and abundance, following a method established by the European Commission (Regulation (EU) 2024/1991 (https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=OJ:L_202401991), with the method specified in (EU) 2025/2188 (https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=OJ:L_202502188)). Moths are one of four pollinator groups covered, alongside bees, bumblebees, butterflies, and hoverflies.

The methodology for moth monitoring has been developed step by step through earlier Swedish pilot studies and the European SPRING project, but several practical questions remain unanswered for Swedish conditions:

- Which type of trap works best?
- What light intensity should be used?
- Where within a survey square should traps be placed?
- How should the surrounding habitat be documented?

Sweden needs answers to these questions in order to adapt monitoring ahead of it starting in earnest from 2027. That's what this pilot project during 2026 aims to provide the basis for.

The two parts of the pilot

Landscape effects (Lund and Uppsala)

Four 1×1 km survey squares, two in Uppland and two in Skåne. Each square contains a regular 3×3 grid with nine trap positions, sampled weekly during the season as far as possible.

This part is intended to help answer:

- How much do moth communities vary within one and the same survey square?
- How does species composition and abundance differ between different parts of the same square?
- How representative is a single trap position for the surrounding environment?

- What can this teach us about where future monitoring stations should place their traps?

The habitat around each trap position is documented in order to link the results to local environmental conditions.

Latitude gradient (Lund–Abisko)

A number of sites along a gradient from southern to northern Sweden, where several different types of light-bucket traps are tested in parallel at each site (see Trap types (/nattfjarilar-manual/en/falltyper/oversikt.html)).

This part is intended to help answer:

- How do different trap types differ in what they catch?
- What role does light intensity play?
- How are catch results affected by latitude and light conditions?
- Does the methodology work equally well across the whole country?

Northern Sweden gets extra attention in this part, since light summer nights can affect how effective light traps are — something particularly relevant for Swedish conditions compared with countries further south in Europe.

Why these tools (ButterflyCount / butterfly-monitoring.net)?

We use the ButterflyCount app and butterfly-monitoring.net because they:

- Integrate image recognition (species identification from photos)
- Have built-in validation after the fact
- Are the most complete tools currently available for this type of survey

Future plan: the goal is to eventually be able to sync data automatically to Artportalen with no extra work for you as a participant. This isn't yet in place — it requires both your consent and approval from Artdatabanken. Even now, though, you can always export your data and manually import it into Artportalen if you'd like it there too.

What happens to the results?

After the season, the results are compiled and analysed to provide a basis for how moth monitoring should be carried out in Sweden as part of the EU's pollinator monitoring (EU-

PoMS) from 2027 onwards. A preliminary report is submitted to Naturvårdsverket at the end of 2026, with a final report in early 2027.

Consent: sharing of contact details

We will partly send out information via newsletter and email — if you don't want your email address to be visible there, just let us know at nattflyn@gmail.com (mailto:nattflyn@gmail.com). To make it possible to quickly exchange experiences and get help with support, we will also have a WhatsApp group, but participation there is voluntary. Your contact details may therefore be shared with other participants in the project via these two channels, but your email can be anonymised and the WhatsApp group is optional.